Collection round 3: recovered instruments, full-text re-reads and supplementary fields (v5)

- Summary
- Full-text re-reads of truncated instruments
- Supplementary fields: whom a rule binds, when it applies, what a standard's legal effect is, when the robot provision arrived
- Second precision audit
- The governance gap, split by collection source
- What round 3 did not do

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Companion to the analysis report. Round 3 executed the items of the round-3 plan that an agent can do alone: re-fetching blocked and missing instruments by official alternative routes, a harvest with service-robot terms, a full-text re-read of truncated statutes, a supplementary schema (whom an instrument binds, dates, legal effect of standards, provision dating), a second precision audit, and a source split of the governance-gap index. Items that need a person are listed in HUMAN_ASSISTANCE.md.

Summary

Round 3 added 160 usable documents to the collection: 108 round-2 candidates whose text could now be read (80 of them principal binding embodied instruments) and 52 new documents, 9 of them reference-list instruments fetched from their titles and the rest from the harvest with service-robot, care, exoskeleton, humanoid and collaborative-robot terms and EuroVoc concepts. The harvest itself added little: of its 43 usable documents only 8 fall in the embodied set and 8 are principal instruments, so the open legal databases hold few service-robot instruments that rounds 1 and 2 had not already found, and the gains of round 3 come from reading texts that were already known. Of the reference-list instruments, 4 are now principal embodied instruments in the collection. The collection now holds 3,320 usable documents, 1,704 in the embodied set, 598 principal embodied instruments and 278 principal binding ones (v4: 3,165 / 1,603 / 505 / 195). Of the 206 round-3 targets, the collectors saved a text for 121, found 74 blocked at official hosts, judged 9 irrelevant and could not locate 2 (Table 80).

The collectors' own reports, with the alternative routes that worked (Publications Office CELEX resolver for EUR-Lex, state legislatures' PDF endpoints, Riigi Teataja and legislation.govt.nz rendered pages, the Kremlin acts bank, Gazzetta Ufficiale issue PDFs) and the metadata corrections they noticed (act numbers, repeal dates, mislabelled gazette copies), are in collection_round3/status/report_*.md.

Table 80. Round-3 collector outcomes for the 206 targets (189 verified-but-unread binding candidates from round 2 and 17 reference-list instruments missing on the strict test)

outcome	targets
saved	113
blocked	74
irrelevant	9
saved_partial	8
not_found	2
not attempted	0

The blocked targets are the manual-download list for a person with ordinary browser access (Table 81).

Table 81. Targets whose official hosts refuse the identified research crawler: the list for manual download

candidate_id	title_en	jurisdiction	source_url	not
EN-014	Ordinance No. 2016-1057 of 3 August 2016 on the experimentation of vehicles with delegated driving on public roads	FR	https://www.legifrance.gouv.fr/eli/ordonnance/2016/8/3/2016-1057/jo/texte	leg oth
EN-015	Decree No. 2018-211 of 28 March 2018 on the experimentation of vehicles with delegated driving on public roads	FR	https://www.legifrance.gouv.fr/eli/decret/2018/3/28/2018-211/jo/texte	leg oth
EN-016	Law No. 2019-1428 of 24 December 2019 on mobility orientation (LOM) – articles 31–32 on autonomous and connected vehicles	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000039666574	leg oth

candidate_id	title_en	jurisdiction	source_url	no
EN-018	Decree No. 2021-873 of 29 June 2021 implementing Ordinance No. 2021-443 on criminal liability and conditions of use of vehicles with delegated driving	FR	https://www.legifrance.gouv.fr/eli/decret/2021/6/29/2021-873/jo/texte	leg oth
EN-019	Law No. 2016-1428 of 24 October 2016 on strengthening the safety of civil drone use	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000033293745	leg oth
EN-038	Royal Decree of 10 April 2016 on the use of remotely piloted aircraft in Belgian airspace	BE	https://www.ejustice.just.fgov.be/eli/arrete/2016/04/10/2016014132/justel	Th nu Mo the del tel
EN-050	Law No. 2015-992 of 17 August 2015 on energy transition for green growth – article 37 IX (authorisation to legislate by ordinance on vehicles with delegated driving)	FR	https://www.legifrance.gouv.fr/loda/id/JORFTEXT000031044385/	leg oth 20
EN-051	Order of 17 April 2018 on the experimentation of vehicles with delegated driving on public roads	FR	https://www.legifrance.gouv.fr/loda/id/JORFTEXT000036868691/	leg oth
EN-052	Law No. 2019-486 of 22 May 2019 on business growth and transformation (PACTE) – article 125 on automated vehicle experimentation	FR	https://www.legifrance.gouv.fr/jorf/article_jo/JORFARTI000038496320	leg oth
EN-053	Ordinance No. 2021-442 of 14 April 2021 on access to vehicle data	FR	https://www.legifrance.gouv.fr/loda/id/JORFTEXT000043370884	leg oth
EN-054	Decree No. 2022-1034 of 21 July 2022 publishing the amendment to the 1968 Vienna Convention on Road Traffic (automated driving systems)	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046081252	leg oth Vie
EN-055	Order of 2 August 2022 on the qualification of remote operators of automated road transport systems (art. R. 3152-3 Code des transports)	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046151685	leg oth
EN-056	Order of 2 August 2022 on the approval procedure for qualified bodies assessing automated road transport systems (art. R. 3152-30 Code des transports)	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046216833	leg oth

candidate_id	title_en	jurisdiction	source_url	not
EN-057	Order of 8 December 2022 on the mandatory consumer information about conditions of use of a vehicle's automated driving system	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000046828949	leg oth
EN-058	Decree No. 2023-644 of 20 July 2023 on access to certain vehicle data for accident prevention, road infrastructure mapping and traffic knowledge	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000047867363	leg oth
EN-059	Decree No. 2024-1063 of 25 November 2024 on automated road freight transport	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000050662872	leg oth
EN-060	Order of 15 April 2025 on operational safety of automated road transport systems using fully automated vehicles for public passenger transport	FR	https://www.legifrance.gouv.fr/jorf/id/JORFTEXT000051484894	leg oth
EN-084	Decree-Law No. 58/2018 of 23 July – registration and compulsory civil liability insurance regime for unmanned aircraft (drones)	PT	https://diariodarepublica.pt/dr/detalhe/decreto-lei/58-2018-115898563	dia file op 20
EN-085	ANAC Regulation No. 1093/2016 of 14 December – operating conditions for use of airspace by civil remotely piloted aircraft systems (drones)	PT	https://diariodarepublica.pt/dr/detalhe/regulamento/1093-2016-105367104	dia file op bot
EU-130	Act on the testing of self-driving vehicles (Norway)	NO	https://lovdata.no/dokument/NL/lov/2017-12-15-112	lov luft sh
EU-131	Regulation on the testing of self-driving motor vehicles (Norway)	NO	https://lovdata.no/dokument/SF/forskrift/2017-12-19-2240	lov luft sh
EU-132	Regulation on unmanned aircraft (Norway, BSL A 7-2)	NO	https://lovdata.no/dokument/SF/forskrift/2024-11-01-2777	lov luft sh
EU-133	Regulation on aircraft without a pilot on board etc. (Norway, 2015)	NO	https://lovdata.no/dokument/SF/forskrift/2015-11-30-1404	lov luft sh
EU-134	Regulation on U-space and an integrated digital aeronautical publication (Norway, BSL A 7-3)	NO	https://lovdata.no/dokument/SF/forskrift/2024-10-23-2572	lov luft sh
AP-002	Drone (Amendment) Rules, 2022 (G.S.R. 108(E))	IN	https://www.civilaviation.gov.in/sites/default/files/2023-10/Drone%20%28Amendment%29%20Rules%2C%202022%20dated%2011%20Feb%202022_0.pdf	civ an blo htt dat
AP-009	Conditional exemption to Government of Telangana for drone operations (MoCA order, 4 August 2021)	IN	https://www.civilaviation.gov.in/node/3286	civ to ww exe arc do

candidate_id	title_en	jurisdiction	source_url	not
AP-087	Motor Vehicles (Trials of Automotive Technologies) Amendment Act 2016 (SA)	AU	https://www.legislation.sa.gov.au/_legislation-documents/lz/v/a/2016/motor-vehicles-trials-of-new-automotive-technologies-amendment-act-2016_10/2016.10.un.pdf	leg an 20 exi
AP-088	Transport Legislation Amendment (Automated Vehicle Trials and Innovation) Act 2017 No 22 (NSW)	AU	https://legislation.nsw.gov.au/view/pdf/asmade/act-2017-41	leg PD Ho Au Tra
AP-089	Road Safety Amendment (Automated Vehicles) Act 2018 (Vic)	AU	https://www.legislation.vic.gov.au/as-made/acts/road-safety-amendment-automated-vehicles-act-2018	leg co link A h
EN-086	Turkish DGCA Instruction on Unmanned Aircraft Systems (SHT-UAV) and UAV registration system	TR	https://web.shgm.gov.tr/documents/sivilhavacilik/files/mevzuat/SHT-IHA_Rev-05.pdf	TL: att po htt pa (re me
IG-112	DoD Directive 3000.09, Autonomy in Weapon Systems (updated 25 January 2023)	US	https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodd/300009p.pdf	es oth
ML-032	Federal Decree-Law No. (26) of 2022 Concerning the Regulation of Unmanned Aircraft	AE	https://uaelegislation.gov.ae/en/legislations/1587	ua /d is
ML-033	Cabinet Resolution No. (110) of 2023 Concerning the UAE Scheme for Drone Products and Systems (UAE Regulation for Unmanned Aircraft)	AE	https://uaelegislation.gov.ae/en/legislations/2177	sa
ML-041	GACAR Part 107 - Operation of Unmanned Aircraft Systems (Version 4)	SA	https://gaca.gov.sa/-/media/Files/PDF/LawsAndRegulation/Aviation-Safety-and-Environmental-Sustainability/GACAR-Safety-Regulations/107-v4.pdf	ww sa htt En sm
ML-096	DAN 151 - Operations of remotely piloted aircraft (RPAS) in matters of public interest	CL	https://www.dgac.gob.cl/	dg (ht am sol 20 fro
US-086	California Vehicle Code Section 38750: Autonomous vehicles (SB 1298, 2012)	US	https://leginfo.legislature.ca.gov/faces/codes_displaySection.xhtml?lawCode=VEH&sectionNum=38750.	leg twi sa
US-108	Indiana Code 9-21-11.5: Personal Delivery Devices (HEA 1072)	US	https://iga.in.gov/laws/2024/ic/titles/9#9-21-11.5	iga Hu
US-109	Indiana Code 35-33-5-9: Warrant requirement for unmanned aerial vehicles (HEA 1009, 2014)	US	https://iga.in.gov/laws/2024/ic/titles/35#35-33-5-9	iga
US-122	Michigan SAVE Act package (Public Acts 332-335 of 2016; MCL 257.665 et seq.): Automated motor vehicles	US	https://www.legislature.mi.gov/Laws/MCL?objectName=mcl-257-665	leg an
US-123	Michigan Unmanned Aircraft Systems Act (Public Act 436 of 2016; MCL 259.301-259.335)	US	https://www.legislature.mi.gov/Laws/MCL?objectName=mcl-Act-436-of-2016	leg

candidate_id	title_en	jurisdiction	source_url	no
US-134	New Hampshire RSA 265:163: Personal delivery devices and mobile carriers (HB 116, 2022)	US	https://gc.nh.gov/rsa/html/XXI/265/265-163.htm	gc. (M
US-182	Wisconsin Executive Order #245 (2017): Governor's Steering Committee on Autonomous and Connected Vehicle Testing and Deployment	US	https://docs.legis.wisconsin.gov/code/executive_orders/2011_scott_walker/2017-245.pdf	do the
REF-06	ISO/TS 15066:2016 collaborative robots	INT	https://www.iso.org/standard/62996.html	iso hu
REF-08	ISO 26262 road vehicle functional safety	INT	https://www.iso.org/standard/68383.html	iso
REF-09	ISO 21448 (SOTIF)	INT	https://www.iso.org/standard/77490.html	iso
REF-12	ANSI/RIA R15.06 industrial robot safety	US	https://webstore.ansi.org/standards/ria/ansiriar15062012	a3 AN
REF-13	ANSI/A3 R15.08 industrial mobile robots	US	https://webstore.ansi.org/standards/ria/ansia3r15082020	a3
AP-020	Unmanned Aircraft (Public Safety and Security) Act 2015 (Act 16 of 2015)	SG	https://sso.agc.gov.sg/Acts-Supp/16-2015	ss
AP-036	CAAT Announcement No. 3 B.E. 2568 (2025) on rules for agricultural drone operations in border/security areas	TH	https://www.caat.or.th/caat-media/167323/	ca po an ap
AS-015	Tianjin Municipal Regulations on Promoting the Development of Intelligent Connected Vehicles	CN	https://gyxxh.tj.gov.cn/ZWGK4147/ZCWJ6355/sjwj/202512/t20251204_7191311.html	tj.g Alt
AS-016	Yangquan Municipal Measures for the Administration of Intelligent Connected Vehicles	CN	https://flk.npc.gov.cn/detail?id=ff8081818c24df58018cce7f027466b2	flk. co
AS-017	Hefei Municipal Regulations on Promoting the Application of Intelligent Connected Vehicles	CN	http://www.ahrd.gov.cn/article.jsp?strId=7a876823fe4544c1a9850923d460665a&strColId=44861419b53244f8910cd00be098270d&strWebSiteId=1448865560847002	An fro
AS-018	Wuhan Municipal Regulations on Promoting the Development of Intelligent Connected Vehicles	CN	https://flk.npc.gov.cn/detail?id=ff808181927f1276019487a8b5db616c	flk res tex
AS-019	Guangzhou Municipal Regulations on Innovative Development of Intelligent Connected Vehicles	CN	https://flk.npc.gov.cn/detail?id=ff808181927f0e7b0194de2edeb60a92	flk

candidate_id	title_en	jurisdiction	source_url	no
AS-020	Ordos Municipal Regulations on Promoting the Development of Intelligent Connected Vehicles	CN	https://flk.npc.gov.cn/detail?id=ff8081819aedd63e019b495526357047	flk co the
AS-023	Changsha Municipal Regulations on the Development of Autonomous Vehicles	CN	https://flk.npc.gov.cn/detail?id=ff8081819bdff392019be3364efd0ab4	flk dat
AS-026	Huzhou Municipal Regulations on Innovative Development of the Intelligent Connected Vehicle Industry	CN	https://flk.npc.gov.cn/detail?id=ff8081819f12cdb7019f3bae7cbf4adc	flk
AS-027	Jiaxing Municipal Regulations on Promoting Road Testing and Application of Intelligent Connected Vehicles	CN	https://flk.npc.gov.cn/detail?id=ff8081819dba1254019dce0118552f83	flk
AS-031	Guangzhou Municipal Regulations on the Development of the Low-Altitude Economy	CN	https://ghzyj.gz.gov.cn/zwgk/newzcfg/ywly/content/post_10120201.html	gz. htt
AS-032	Zhuhai Special Economic Zone Regulations on Low-Altitude Transport Construction and Management	CN	https://flk.npc.gov.cn/detail?id=ff808181927f093101939b3d540500a6	flk
AS-036	Zhejiang Provincial Provisions on Public Security Administration of Unmanned Aircraft	CN	https://flk.npc.gov.cn/detail?id=2c90e5ba6a291d8f016a4cc775c3034a	flk
AS-038	Zhejiang Provincial Civil Aviation Regulations	CN	https://flk.npc.gov.cn/detail?id=ff8081819f12cdb7019f369a9f932c52	flk
AS-104	Enforcement Rule of the Autonomous Vehicles Act	KR	https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=271493	law
AS-106	Rules on Performance and Standards of Motor Vehicles and Parts (Motor Vehicle Safety Standards)	KR	https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=288083	law
AS-112	Enforcement Rule of the Aviation Safety Act	KR	https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=287951	law an
AS-114	Act on Promotion of Development and Commercialization of Autonomous Ships	KR	https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=281889	law (en
AS-115	Rules on Occupational Safety and Health Standards	KR	https://www.law.go.kr/LSW/lsInfoP.do?lsiSeq=273603	law rul
AS-121	Regulations on Issuance of Plates for Unmanned Vehicle Innovative Experimentation Projects	TW	https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090084	law mo
AS-123	Civil Aviation Act (Chapter 9-2 on Remotely Piloted Aircraft)	TW	https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090001	law a=

candidate_id	title_en	jurisdiction	source_url	no
AS-125	Rules for Investigation of Major Flight Accidents Involving Remotely Piloted Aircraft	TW	https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090086	law
AS-126	Regulations on Delegation of Drone Inspection and Operator Testing	TW	https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0090085	law
AS-127	Regulations on Application and Management of Drone Flights within Designated Commercial Port Areas	TW	https://law.moj.gov.tw/LawClass/LawAll.aspx?pcode=K0080041	law
AS-134	Notice of Beijing Municipal Government Publishing the Scope of Controlled Airspace for Unmanned Aircraft	CN	https://www.beijing.gov.cn/zhengce/zhengcefagui/202508/t20250804_4165802.html	No (W (ab in t
REF-16	Taiwan Industrial Robot Hazard Prevention Standards (1987, amended 2018)	TW	https://law.moj.gov.tw/LawClass/LawSearchContent.aspx?kw=工業用機器人危害預防標準	law En

Full-text re-reads of truncated instruments

The excerpt policy stops at 110,000 characters; 104 principal embodied instruments exceeded it. Re-reading them with up to 400,000 characters under the same prompt left the document class unchanged in 96.2% and the relevance rating in 87.5%; principle levels agreed exactly in 89.7% of field-document pairs, and the full text raised a level in 6.4% of pairs (Table 74). The full-text labels replace the excerpt labels for these documents. The documents whose set-defining labels changed are listed in Table 75.

Table 74. Full-text re-read (up to 400,000 characters) against the excerpt-based label (110,000 characters) for the 104 principal instruments whose text exceeded the excerpt: agreement per field

field	n	agree	agree_pct
doc_class	104	100	96.2
binding_nature	104	101	97.1
instrument_status	104	100	96.2
ai_scope	104	93	89.4
embodiment_status	104	96	92.3
relevance_to_embodied_ai_governance	104	91	87.5
page_quality	104	96	92.3
primary_embodied_category	104	93	89.4
risk_framing	104	90	86.5
embodied_categories (mean Jaccard)	104	88.7	85.3
governance_topics (mean Jaccard)	104	79.8	76.7
governance_mechanisms (mean Jaccard)	104	82.6	79.4
application_sectors (mean Jaccard)	104	89.4	86
principles (exact level, all 20)	2080	1865	89.7
principle levels raised by the full text	2080	134	6.4
principle levels lowered by the full text	2080	81	3.9

Table 75. Truncated principal instruments whose set-defining labels changed when the full text was read

doc_id	title	changed_fields	excerpt_class	fulltext_class	excerpt_relevance	fulltext_relevance
R2-7bb2229f	Regulation (EU) 2023/1230 of the European Parliament and of the Council of 14 June 2023 on machinery and repealing Directive 2006/42/EC and Council Directive 73/361/EEC	ai_scope	governance_instrument_binding	governance_instrument_binding	3	3
R2-8991668e	Official Mexican Standard NOM-107-SCT3-2019, establishing requirements to operate a remotely piloted aircraft system (RPAS) in Mexican airspace	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	3	2
R2-6da0ddec	Royal Decree 517/2024 of 4 June developing the legal regime for the civil use of unmanned aircraft systems (UAS)	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	2	3
R2-82f8813a	Civil Aviation Regulations 2016	ai_scope	governance_instrument_binding	governance_instrument_binding	1	1
R2-f67080c2	Standards Document: Operation of Aircraft, Part IV – International Operations – Remotely Piloted Aircraft Systems (RPAS)	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	3	2
D1315	White Paper of the Austrian Council on Robotics and Artificial Intelligence	relevance_to_embodied_ai_governance	governance_proposal_or_draft	governance_proposal_or_draft	2	3
R2-c5eb5314	Civil Aviation Safety Regulations 1998	ai_scope	governance_instrument_binding	governance_instrument_binding	2	2
D1342	National AI Strategy	ai_scope	governance_instrument_soft	governance_instrument_soft	1	1

doc_id	title	changed_fields	excerpt_class	fulltext_class	excerpt_relevance	fulltext_relevance
D1290	Proposal for a Regulation of the European Parliament and of the Council Laying Down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts	binding_nature	governance_proposal_or_draft	governance_proposal_or_draft	2	2
R2-f57251ed	Ordinance on Industrial Safety and Health	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	2	1
D1374	Proposals for an Italian Strategy for Artificial Intelligence	doc_class; relevance_to_embodied_ai_governance	governance_instrument_soft	governance_proposal_or_draft	1	2
R2-6ecde273	Decision No. 2021-817 DC of 20 May 2021 - Global Security Law Preserving Freedoms	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	2	1
R2-f48a8103	Proposal for a Regulation of the European Parliament and of the Council on machinery products	binding_nature; ai_scope	governance_proposal_or_draft	governance_proposal_or_draft	3	3
R2-9c4dc434	Civil Aviation Law	ai_scope	governance_instrument_binding	governance_instrument_binding	1	1
R2-1617d407	Civil Aviation Directive 6011 Part (V) – Unmanned Aircraft System: Special UAS Project	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	3	2
D1186	Hungary's Artificial Intelligence Strategy 2020–2030	relevance_to_embodied_ai_governance	governance_instrument_soft	governance_instrument_soft	1	2
R2-7115e06c	Opinion on the Integration of Autonomy into Lethal Weapon Systems	doc_class	governance_instrument_soft	government_report_or_parliamentary	3	3

doc_id	title	changed_fields	excerpt_class	fulltext_class	excerpt_relevance	fulltext_relevance
R2-92b0acef	Proposal for a Regulation amending Regulations (EU) 2017/745 and (EU) 2017/746 as regards simplifying and reducing the burden of the rules on medical devices and in vitro diagnostic medical devices, and amending Regulations (EU) 2022/123 and (EU) 2024/1689	binding_nature; relevance_to_embodied_ai_governance	governance_proposal_or_draft	governance_proposal_or_draft	2	1
R2-2b6aea0b	Ordinance on Type Designation of Devices (Road Vehicles)	ai_scope	governance_instrument_binding	governance_instrument_binding	2	2
R2-b70da4e3	Our Common Agenda Policy Brief 9: A New Agenda for Peace	doc_class; relevance_to_embodied_ai_governance	governance_instrument_soft	governance_proposal_or_draft	2	1
R2-80171de4	Directive 2006/42/EC of the European Parliament and of the Council of 17 May 2006 on machinery, and amending Directive 95/16/EC (recast)	ai_scope	governance_instrument_binding	governance_instrument_binding	2	2
R2-a586cca8	Regulation No 79 of the Economic Commission for Europe of the United Nations (UN/ECE) — Uniform provisions concerning the approval of vehicles with regard to steering equipment	relevance_to_embodied_ai_governance	governance_instrument_binding	governance_instrument_binding	2	3
R2-35e40c62	Adra Strategic Research, Innovation, and Deployment Agenda 2025-2027	doc_class	governance_proposal_or_draft	governance_instrument_soft	2	2
R2-431b7f3a	Ministerial Ordinance Establishing Technical Standards for Railways	ai_scope	governance_instrument_binding	governance_instrument_binding	1	1

doc_id	title	changed_fields	excerpt_class	fulltext_class	excerpt_relevance	fulltext_relevance
R2-322c6200	The Workplace Safety and Health Act — Workplace Safety and Health Regulation, Manitoba Regulation 217/2006	relevance_to_embodied_ai_governance; ai_scope	governance_instrument_binding	governance_instrument_binding	2	1
D0104	Ethics and governance of artificial intelligence for health: WHO guidance	ai_scope	governance_instrument_soft	governance_instrument_soft	1	1

Supplementary fields: whom a rule binds, when it applies, what a standard's legal effect is, when the robot provision arrived

A second, smaller annotation pass asked five questions the codebook had not asked, of every principal embodied instrument, with a quote required for every answer. Among the 278 principal binding instruments the text names whom the rule binds in 267 cases (operators or deployers and public bodies most often; Table 71), an entry-into-force date in 181, a later application date in 49 and a repeal or expiry date in 15 (Table 70). The year the robot-relevant provision arrived could be given for 226 binding instruments, but mostly because the whole instrument is about such systems (182 cases), so the provision year is the document year; an amendment history in the text dated the provision in 26 cases and the provision itself stated its date in 16; in 54 the year could not be determined from the captured text. Only 7 binding instruments carry a provision year later than the document's recorded year (Table 73). The consolidated statutes dated before 2010 are the case that matters for trend analyses, and for 15 of the 16 the captured consolidation carries no amendment history, so the model could not date their automated-driving or drone articles: these need a person with the amendment record (HUMAN_ASSISTANCE.md, the legal-check item). Among the technical standards, 45 state on their public page that they are voluntary and 1 that they are harmonised or designated under a law (Table 72).

Table 70. Supplementary fields (v5) stated in the text: counts of principal instruments for which the model found each field in the document, and the number whose robot-relevant provision is dated later than the document's own year

population	n	bound_parties_stated	entry_into_force_date	applicability_date	repeal_or_expiry_date	robot_provision_year	provision_year_later_than_document_year
principal embodied instruments	598	436	287	64	36	526	8
principal binding embodied instruments	278	267	181	49	15	226	7
principal binding, dated before 2010	16	16	9	2	0	1	1

Table 71. Whom the 278 principal binding embodied instruments bind, as stated in their text (a document may bind several parties)

party	binding_instruments	share_pct
public_bodies_or_government	176	63.3
operators_or_deployers	171	61.5
manufacturers_or_producers	108	38.8
owners_or_users	106	38.1
pilots_or_drivers	70	25.2
member_states_or_subnational_governments	68	24.5
individuals_or_general_public	51	18.3
certification_or_notified_bodies	39	14
importers_or_distributors	38	13.7
providers_or_developers	30	10.8
employers	16	5.8
none_stated	9	3.2
researchers_or_institutions	7	2.5
not_applicable	2	0.7

Table 72. Legal effect of the 61 principal instruments annotated as technical standards, as stated on the captured text

legal_effect	standards	share_pct
voluntary	45	73.8
unknown	15	24.6
harmonised_or_designated	1	1.6

Table 73. Principal binding embodied instruments whose robot-relevant provision was added later than the document's recorded year, according to the amendment history in the text (the dates to use for provision-level trend analyses)

doc_id	title_en	jurisdiction_country	year_final	robot_provision_year	robot_provision_basis
R2-aa8d88b9	Ordinance on Transport Safety in the Freight Motor Vehicle Transport Business	JP	1990	2025	amendment_history_in_text
R2-ee56555b	Traffic Act (Estonia)	EE	2010	2017	amendment_history_in_text
R2-5d2b19e5	Finnish Aviation Act 864/2014 (consolidated version)	FI	2014	2020	amendment_history_in_text
R2-ce2171ab	Ordinance for Enforcement of the National Strategic Special Zones Act (MLIT-related)	JP	2014	2022	amendment_history_in_text
R2-f8592e32	The Air Navigation Order 2016	GB	2016	2025	amendment_history_in_text
R2-295bf91b	Air Navigation (Aircraft Noise) Regulations 2018	AU	2018	2021	amendment_history_in_text
R2-be225994	Regulations Amending the Canadian Aviation Regulations (Remotely Piloted Aircraft Systems): SOR/2019-11	CA	2018	2019	whole_instrument_is_about_such_systems

Second precision audit

An independent agent re-read a random sample of 100 of the 195 principal binding embodied instruments of v4 (seed 2027) against the five criteria of the v3 audit. Shares: distinct instrument 85%, binding 99%, substantive embodied provisions 76%, autonomy or AI content 73%, explicit AI 9%; all three main criteria jointly 73% (Wilson 64 to 81), implying about 142 genuine instruments (124 to 157) in the set of 195. The v3 audit of the earlier set of 299 found 54% (44 to 63). The sample was drawn from the v4 set before round 3; applied to the v5 set of 278 principal binding instruments the same share implies about 203 (177 to 224). The failures are mostly binding instruments without a substantive embodied provision (a drone or robot named only in a list of functions or definitions), then register, tracking and catalogue pages and date-only amendments that the counting-unit rules let through. The audit report with the failing documents is review/v5_audit/precision_audit_v5.md.

The governance gap, split by collection source

Because round 2 searched legal databases for drones, vehicles and machinery, the instrument side of the governance-gap index could inherit the search terms. Table 7b computes the index separately against the round-1 instruments (web search of robot-specific and general-AI sources) and against the round-2 and round-3 instruments (database harvests and recovered texts). The gap is positive against both sources for social companion robots (+35 and +41 points), humanoid general purpose (+6 and +20 points), educational robots (+5 and +6 points), sex robots (+3 and +4 points), other embodied (+3 and +4 points): research attention to these machine classes exceeds their share of instruments whichever way the instruments were found, so the headline gap for service, companion and humanoid robots is not an artefact of the round-2 terms. The instrument surplus is present against both sources for drones uas (-31 and -38), autonomous road vehicles (-55 and -26), mobile industrial logistics (-16 and -5), agricultural robots (-18 and -3), autonomous maritime or rail (-6 and -4), rehabilitation exoskeleton prosthetic (-2 and -2). The gap appears against only one source for care assistive robots (+0 against round-1 instruments, +21 against harvested ones): for these classes the round-1 robot-specific sources supplied standards and guidance that the database harvests did not, so the size of the gap depends on how instruments were found. The sign flips for surgical medical robots (-8 against round-1 instruments, +10 against harvested ones), professional service robots (-15 against round-1 instruments, +10 against harvested ones), collaborative robots (-16 against round-1 instruments, +3 against harvested ones), industrial manipulators (-41 against round-1 instruments, +3 against harvested ones): the round-1 instrument set, drawn from robot-specific sources, is rich in standards for these classes, and the harvested set is not, so for them the gap depends on the source and should not be read as a finding.

Table 07b. Governance-gap index by collection source: research share (n=509) minus the principal-instrument share among round-1 instruments (web search, n=59) and among round-2 instruments (database harvest, n=539); a gap present in both columns does not depend on the round-2 search terms

category	level1	research_pct	n_research	instrument_pct_round1	n_instruments_round1	gap_round1	instrument_pct_round2	n_instruments_ro
social_companion_robots	Service & social	43	219	8.5	5	34.6	2.4	
care_assistive_robots	Service & social	27.5	140	27.1	16	0.4	6.5	
humanoid_general_purpose	Humanoid & general-purpose	22.8	116	16.9	10	5.8	2.4	
surgical_medical_robots	Medical & body-integrated	14.5	74	22	13	-7.5	4.1	
professional_service_robots	Service & social	17.5	89	32.2	19	-14.7	7.4	
educational_robots	Service & social	6.9	35	1.7	1	5.2	1.1	
domestic_household_robots	Service & social	7.7	39	8.5	5	-0.8	2.8	
sex_robots	Service & social	4.5	23	1.7	1	2.8	0.6	
other_embodied	Other embodied	4.7	24	1.7	1	3	1.1	

category	level1	research_pct	n_research	instrument_pct_round1	n_instruments_round1	gap_round1	instrument_pct_round2	n_instruments_ro
biohybrid_or_cyborg	Medical & body-integrated	2.2	11	0	0	2.2	0.4	
collaborative_robots	Industrial & logistics	6.3	32	22	13	-15.7	3.2	
toys_entertainment_robots	Service & social	1.2	6	0	0	1.2	0.6	
military_autonomous_weapons	Security & military	10.4	53	20.3	12	-9.9	9.1	
space_or_extreme_environment	Vehicles & field	3.1	16	10.2	6	-7	2.6	
industrial_manipulators	Industrial & logistics	10	51	50.8	30	-40.8	7.2	
security_policing_robots	Security & military	3.1	16	6.8	4	-3.6	4.5	
rehabilitation_exoskeleton_prosthetic	Medical & body-integrated	2.8	14	5.1	3	-2.3	5	
autonomous_maritime_or_rail	Vehicles & field	1	5	6.8	4	-5.8	4.6	
agricultural_robots	Vehicles & field	2.4	12	20.3	12	-18	5.4	
mobile_industrial_logistics	Industrial & logistics	3.1	16	18.6	11	-15.5	7.8	
delivery_robots	Industrial & logistics	2.2	11	3.4	2	-1.2	9.6	
autonomous_road_vehicles	Vehicles & field	12.4	63	67.8	40	-55.4	38.4	
drones_uas	Vehicles & field	6.3	32	37.3	22	-31	43.8	

What round 3 did not do

No person labelled any document (the human-validation kit is in data/human_validation/); hosts that refuse the identified crawler were not circumvented (Table 81 lists them for manual download); technical standards are still read from scope pages; and the labels of the new and recovered documents carry the same model limits as the rest.